

Supermarket Edge AI CCTV System

Site Integration & Infrastructure

Deployment Plan

Infrastructure Assessment, Router & PoE Switch Upgrades, UPS Sizing, Camera Specifications, and Optional Accessories

1. 1. Commercial Security Router / Gateway (or use existing)

Model	Ports & Speed	Key Capabilities	Est. Price
TP-Link Omada ER707-M2 (Primary Choice)	2× 2.5GbE RJ45 + 4× 1GbE + 1× SFP	Hardware multi-WAN failover (4G/5G backup), multi-VLAN isolation (CCTV vs POS vs Wi-Fi), WireGuard VPN, 500k sessions	\$165 – \$190 USD
Ubiquiti UniFi Cloud Gateway Max (UCG-Max)	5× 2.5GbE RJ45 ports	Full 2.5 Gbps wire-speed routing with IDS/IPS, integrated UniFi SDN console, advanced traffic filtering	\$199 – \$220 USD

3.2 High-Density PoE+ Managed Switch (Powering 32 Cameras)

Model	Port Count & PoE Budget	Bandwidth & Uplinks	Est. Price
TP-Link Omada TL-SG3452P (Top Recommendation)	48× Gigabit PoE+ ports (384W total budget)	4× Gigabit SFP slots. Powers 32 cameras (@ ~6.5W = 208W, 54% budget) + 16 spare ports for Edge PC, NAS, POS, APs	\$480 – \$520 USD
Dual TP-Link TL-SG3428MP (Alternative 2-Switch)	2× 24× PoE+ ports (384W budget each = 768W)	4× SFP slots. Distributes load across 2 separate 1U switches for physical redundancy	2× \$285 = \$570 USD

4. Power Protection & UPS Sizing Plan (Graceful Auto-Shutdown)

- Edge AI PC (Core i5 + RX 9060 XT 16GB): ~220W – 240W
- 32 PoE Cameras (via PoE Switch): ~190W – 210W (32 × ~6W)
- PoE Switch & Router Logic: ~45W
- Existing NAS Server (4–8 Bays): ~60W – 75W
- Total Operational Power Load: ~515W to 570W

Model	Capacity & Waveform	Runtime & Features	Est. Price
APC Smart-UPS SMT1500IC (Recommended Commercial)	1500VA / 1000W Pure Sine Wave	18–24 mins runtime at 550W. USB signaling, SmartConnect cloud monitoring, AVR line conditioning	\$550 – \$620 USD
CyberPower PR1500ELCD (Cost-Effective Option)	1500VA / 1500W Pure Sine Wave	20–26 mins runtime at 550W. USB + Serial port, GreenPower high-efficiency circuit, LCD diagnostic	\$460 – \$510 USD

5. Camera Specifications Checklist (32 Channels)

To ensure 100% optical performance for both human surveillance and AI computer vision (YOLOv8 person tracking, shelf wrist reach, and theft detection), the cameras are mapped to specific lens focal lengths, sensors, and form factors:

Zone / Dept	Qty	Recommended Model	Optics & Lens	AI & Vision Purpose
Grocery Aisles 1–12	12	Dahua IPC-HDW2441T-S	3.6mm Fixed Lens (88° FOV), 4MP Starlight, 120dB WDR, 30m IR	Long corridor perspective, down-aisle tracking, trolley clash, shelf dwell detection
Fresh Produce & Bakery	4	Dahua IPC-HDW2441T-S	2.8mm Wide Lens (107° FOV), 4MP Starlight, 120dB WDR	Wide department overview, color fidelity for fresh goods, produce display density
POS Checkouts 1–6	6	Dahua IPC-HDW2441T-S	3.6mm or 6mm Narrow Lens, 4MP, 30 FPS, Built-in Mic	Narrow focus on barcode scanner plate and cash drawer for sweet-hearting & queue

wait times

Entrances & Foyers	4	Dahua IPC-HDBW2441E-S	2.8mm Wide Lens, 4MP, True WDR (120dB), IK10 Vandal Dome	Counters bright daylight glare from glass doors; customer entry/exit tripwire counts
High-Value & Logistics	6	Dahua IPC-HDW2441T-S + IPC-HFW2441S-S (Bullet)	3.6mm Lens, 4MP Starlight, Built-in Mic, IP67 Weatherproof	High-theft shelf reach tracking (Liquor & Pharmacy), loading dock logistics, walk-in cold chain

6. Additional Infrastructure & Optional Items Checklist

The following physical hardware items must be verified on site. If missing, they should be procured to complete the installation:

Item Description	Specification / Rating	Category	Est. Price	Status
Server Equipment Cabinet	9U or 12U Wall-Mount Enclosure (600mm Depth, Glass Door, Lockable)	Mandatory	\$140 – \$180	Procure if missing
Cat6 Patch Panel (48-Port)	1U 19" Cat6 110 Punch-Down or Keystone Jack Patch Panel	Mandatory	\$35 – \$50	Procure if missing
Solid Copper Cat6 Cable	305m (1000ft) Box UTP Solid Bare Copper 23AWG (Blue/Grey)	Mandatory	\$110 – \$140 / box	2 boxes (~600m)
Pass-Through RJ45 Plugs	Cat6 Pass-Through RJ45 8P8C Gold-Plated Connectors + Strain Relief Boots (Pack of 100)	Mandatory	\$18 – \$22	Procure
1U Horizontal Cable Duct	19" 1U Finger Duct Cable Management Panel with Cover	Recommended	\$15 – \$20	Procure
Rackmount 8-Outlet PDU	1U 19" Horizontal PDU Power Strip with Surge Suppression (10A/16A)	Mandatory	\$35 – \$45	Procure

Cat6 Molded Patch Leads	0.5m & 1.0m Snagless Cat6 Patch Cords (Pack of 35 for switch-to-panel)	Mandatory	\$28 – \$35	Procure
UPS USB Communication Cable	Type-A to Type-B or RJ50-to-USB Cable for APC/CyberPower signaling	Mandatory	\$10 – \$12	Included in UPS box

7. Complete Turnkey Procurement Budget

Procurement Package	Included Components	Subtotal Range (USD)
1. Edge AI Appliance PC	Intel Core i5, RX 9060 XT 16GB, 16GB DDR5, 512GB NVMe, 650W PSU, Case	\$833 – \$953 USD
2. Network Router & PoE Switch	TP-Link ER707-M2 2.5GbE Router + TL-SG3452P 48-Port PoE+ Switch (384W)	\$645 – \$710 USD
3. Battery Protection (UPS)	APC Smart-UPS 1500VA or CyberPower PR1500ELCD with USB signaling	\$460 – \$620 USD
4. Rack, Cabling & Accessories	9U Wall Rack, 48-port Patch Panel, 2x Cat6 boxes, PDU, Patch cords	\$391 – \$484 USD
5. 32x Cameras (If Procuring New)	32× Dahua WizSense 4MP WDR Eyeball/Dome IP Cameras (@ ~\$75–\$85/unit)	\$2,400 – \$2,720 USD (Optional if existing)

TURNKEY INFRASTRUCTURE (EXCLUDING CAMERAS IF EXISTING): \$2,329 – \$2,767 USD

TOTAL COMPLETE SYSTEM (EDGE PC + ROUTER + SWITCH + UPS + RACK + 32 CAMERAS): \$4,729 – \$5,487 USD